DLMCSPSE01 – Project Software Engineering

Pre-Code Unit Test Case Generator

Phase 2 – Development & Reflection

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Outline

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The Problem

Manual TDD Is Slowing Developers Down

Test-Driven Development requires unit test cases **before** writing any implementation code.

Today, developers create Excel spreadsheets by hand a process that is:

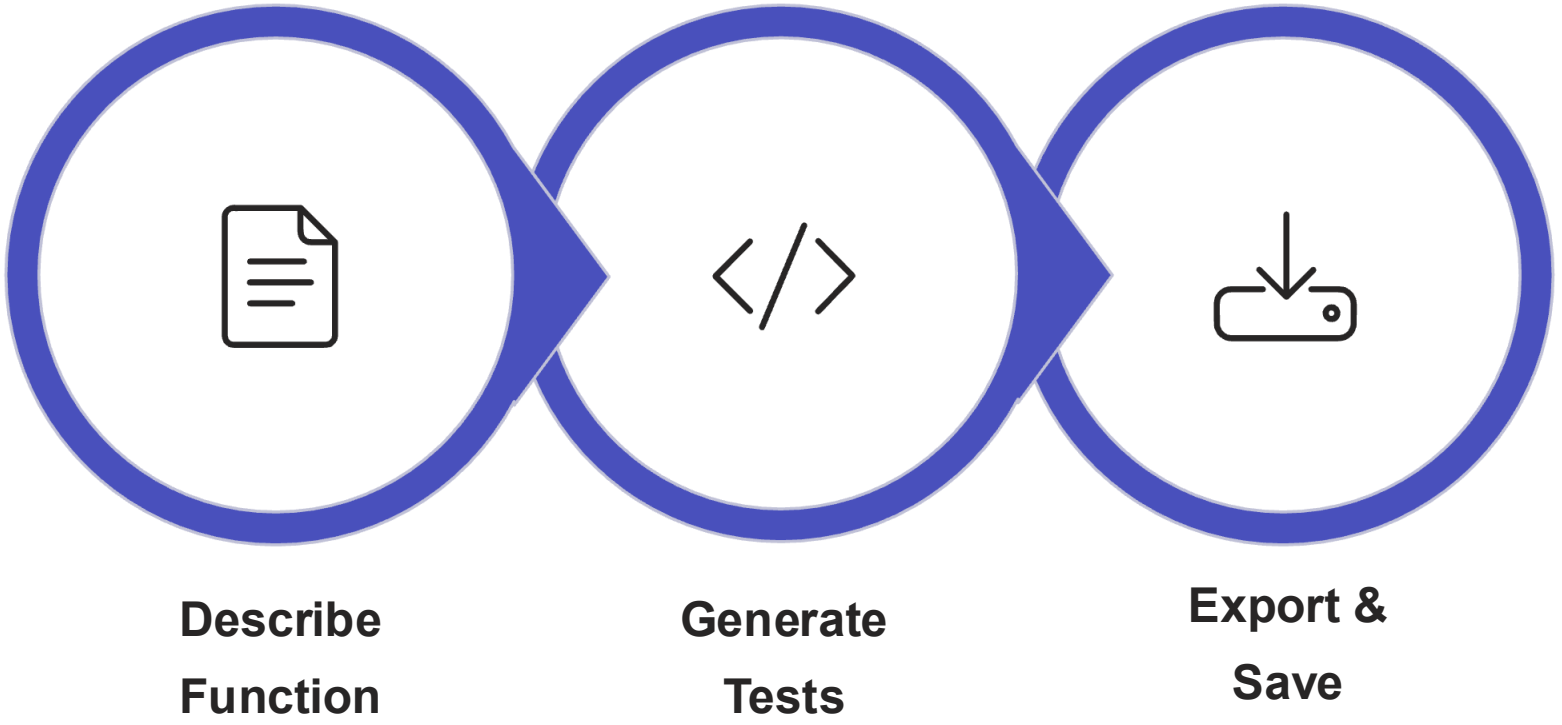
Time-Consuming
Repeated manual effort per function

Error-Prone
Human mistakes in edge case coverage

Inconsistent
No standard format across teams

Proposed Solution

A structured web tool – not a chatbot. The developer fills a form; the AI operates invisibly behind it.



No chat interface, no copy-pasting. Form input → structured data table → one-click .xlsx export

Target Users

- **Software Developers: Pain Point** – Manual Excel test case creation before every function slow
- **QA Engineer: Pain Point –** Inconsistent test case formats across developers
- **Tech Lead: Pain Point –** No standard documentation format within the team

Out of Scope:

- End users / citizens
- Project managers without coding background
- Non-TDD teams: this tool is specifically for developers writing pre-code test documentation

Six Core Features

AI Generation

Gemini API returns strict JSON – 6+ test cases per function across happy path, boundary, negative, and edge categories

Inline Editing

Any generated cell is editable before export

Excel Export

One-click .xlsx via ExcelJS with bold headers and alternating row colours

Auto-Persistence

Every session saved to Supabase PostgreSQL automatically

History Library

Search, filter, reload, and re-download any past session

Structured Tool

AI is invisible – no conversational UI, pure form-to-table design

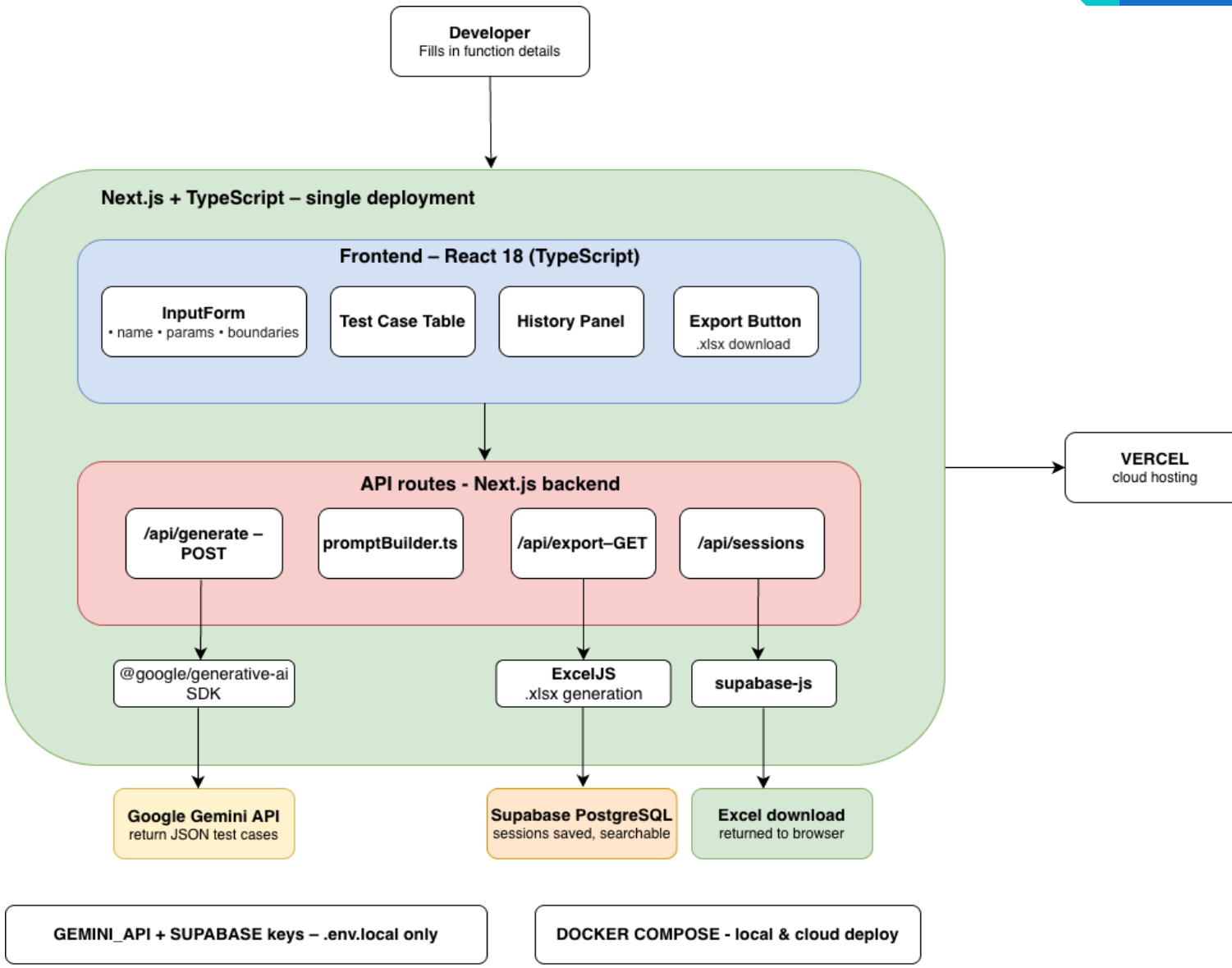
Technology Stack

Technology	Role	Rationale
Next.js 14	Full-stack framework	Single codebase — UI + API routes; no separate server
TypeScript 5	Static type system	Shared interfaces across all layers; compile-time safety
React 18	Frontend UI Library	Input Forms, History, Export Functionality
Google Gemini API	AI structured output	Strict JSON-only responses; no conversational UI
Supabase	Managed PostgreSQL	Persistent session storage; free 500 MB; searchable history
ExcelJS 4	Excel generation	.xlsx export with bold headers and alternating row colors
Docker Compose	Containerization	one-command local deployment
Vercel	Cloud Hosting	GitHub integration; env vars injected

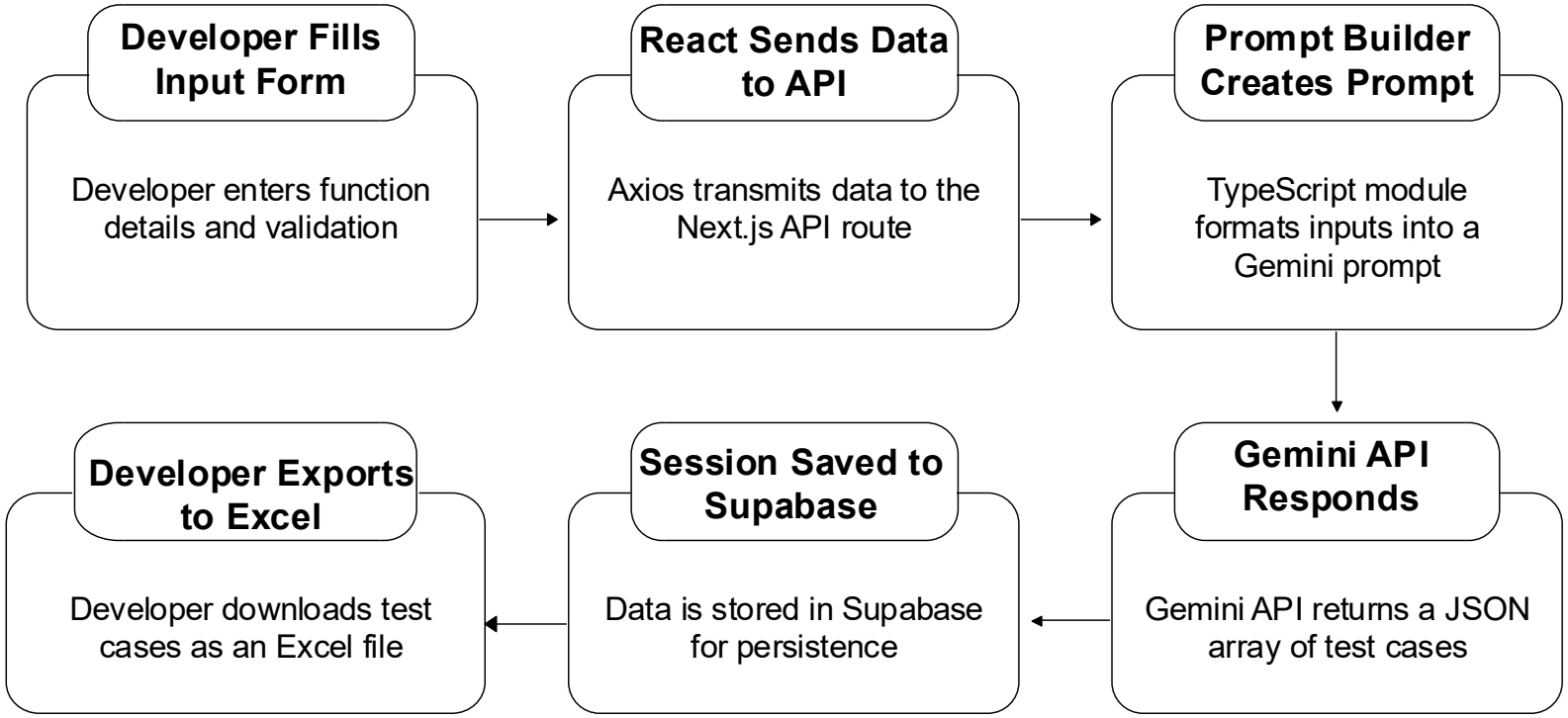
Database Design – Supabase sessions table

Column	Type	Constraint	Purpose
id	UUID	PRIMARY KEY	Auto-generated – gen_random_uuid()
function_name	TEXT	NOT NULL	Searchable – displayed in History Library
created_at	TIMESTAMP	DEFAULT NOW ()	Sort newest-first in session list
form_inputs	JSONB	NOT NULL	Full form data – repopulates InputForm on reload
test_cases	JSONB	NOT NULL	Test case array – repopulates table; used for re-export

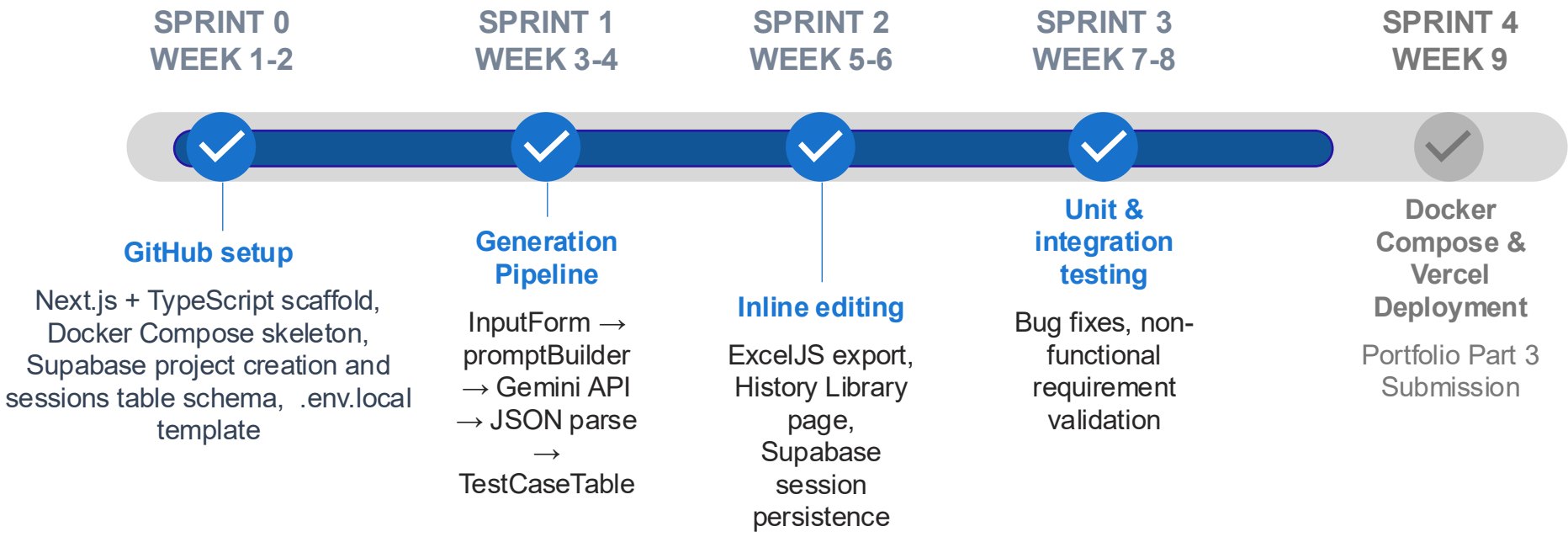
Architecture – System Component Overview



Architecture – Data Flow



Agile/Scrum Sprint Plan



Functional Requirements

- **FR-1:** System generates ≥6 categorized test cases per function submission
- **FR-2:** All sessions persisted to Supabase within 2 s of generation
- **FR-3:** History Library supports search by function name and filter by test category
- **FR-4:** .xlsx export produced with bold headers and alternating row colors

Non – Functional Requirements

- **NFR-1 Performance:** Test case generation completed end-to-end
- **NFR-2 Availability:** ≥99.5% uptime via Vercel + Supabase free tier
- **NFR-3 Reliability:** Session data persisted reliably to Supabase
- **NFR-4 Maintainability:** All source files commented; TypeScript strict mode enforced
- **NFR-5 Scalability:** Backend handles concurrent requests

Risk Register and Mitigation

ID	Risk	Impact	Mitigation Strategy
R-01	Gemini API returns malformed JSON	High	promptBuilder.ts enforces strict JSON-only contract; fallback error boundary in UI
R-02	API rate limit hit (15 req/min)	Low	Exponential retry logic in /api/generate; user-facing error message with retry prompt
R-03	Supabase connection fails on Vercel	Medium	App degrades gracefully – generation still works; save fails silently with user notification
R-04	Excel file corrupt on download	Medium	Content-Type header enforced
R-05	Docker environment drift, Works locally, fails in CI/CD	Medium	Pin all image versions in docker-compose.yml; test container build in GitHub Actions

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Thank You

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